

Digital Twin Model For Simulating Urban Road Traffic Flows

K Kavya

Department of Computer Science and Engineering
Dayananda Sagar University
Bengaluru, India
kkavya0804@gmail.com

Revathi V

Department of Computer Science and Engineering
Dayananda Sagar University
Bengaluru, India
revathi-cse@dsu.edu.in

Nandini K

Department of Computer Science and Engineering
Dayananda Sagar University
Bengaluru, India
nandini-cse@dsu.edu.in

Abstract—Urban traffic congestion is a major issue in smart cities today and it causes longer travel time, more fuel consumption, and pollution in the environment. The current research presents a Digital Twin-based urban traffic simulation framework as a solution to this problem which enables real-time monitoring, analysis, and control of the traffic. The proposed system is based on the SUMO traffic simulator which simulates a four-way signalized intersection and it is integrated with Python through the TraCI interface for establishing real-time bi-directional communication. Parameters such as number of vehicles, average speed and directional flow are being gathered and scrutinized all the time. The adaptive traffic signal control is applied to improve the performance of the intersection depending on real-time congestion situation. The experiments' findings indicate that higher traffic density results in lower vehicle speeds which is a precise depiction of the traffic behavior in the real world. The digital twin excellently reflects the traffic dynamics thus enabling data-driven decision-making and making the traffic flow more efficient. The authors of this paper highlight that the digital twins can be the leading technology for Intelligent Transportation Systems (ITS) and smart city mobility management in the future.

I. INTRODUCTION

The process of urbanization has steadily increased the need for transportation systems that are both effective and efficient and this has resulted in major traffic congestion in cities[1]. The use of traditional traffic management techniques such as fixed-time traffic signals and manual traffic policing is however not sufficient to meet the dynamically changing traffic conditions. This leads to poor signal timing, buildup of longer queues, and consequently, there is an increase in the delays experienced[2].

The rapid growth of digital twin technology has ushered in a new era for the transportation systems of the future. A digital twin refers to a computer-generated mirror image of a physical system that is able to receive data collections in real-time and thus adjusting its behavior accordingly[3]. In the field of traffic management, the use of digital twins means that the constant monitoring, simulation, and optimization of traffic flow can

be done without interfering with the operations of the real world[4].

A Digital Twin-aided traffic control system for an urban intersection with four roads is the main subject of this research. The system employs SUMO as a tool for microscopic traffic simulation and Python-based TraCI control to change the traffic signals dynamically according to the real-time congestion levels. By looking at traffic density and vehicle speed, the suggested system shows how digital twins can enhance traffic efficiency and help in the planning of smart city transports[5].

II. LITERATURE REVIEW

Digital Twin (DT) has become a significant innovation for connecting the physical world with its virtual counterpart. A Digital Twin is a live digital version of a real-world system that through continuous data flow mimics its characteristics, operations, and interactions. NASA was the first to put forward the idea but it has since changed into a versatile solution for various industries like manufacturing, healthcare and intelligent transportation systems [1].

Urban systems that are complex can be monitored, simulated and predicted in real-time due to the combination of various technologies, such as the Internet of Things (IoT), Artificial Intelligence (AI), cloud computing, and data analytics that are considered as enablers [6]. Researchers have looked at the possibility of bringing Digital Twins into the transportation infrastructure in the last few years, with the aim of making traffic flow smoother, the whole system being more energy-efficient, and even supporting smart city planning. Tu et al. have mentioned that the transportation system in the cities can be assessed for energy efficiency in real-time scenario through DT-based models by merging sensor data from the real world and simulation-based predictive models [7].

The study by the authors is about the technological integration where Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA) have been used to measure the performance of road transport infrastructures in different cities

thus showing the possibility of DTs to provide realistic as well as data-driven evaluations. Integrating machine learning with Digital Twins for predicting traffic flow is an important research direction. Real-time data from IoT-enabled sensors is the input to the LSTM (Long Short-Term Memory) neural networks which are most commonly used to predict traffic conditions shortly [8].

The LSTM-DT based frameworks have been shown to predict urban traffic congestion and vehicle flow patterns more accurately than the conventional approaches like ARIMA and SVR (Support Vector Regression) [9]. Such prediction capabilities are very much beneficial for traffic control that is adaptive and for smart city areas determining the best routing strategies. Integrating machine learning with Digital Twins for predicting traffic flow is an important research direction. Real-time data from IoT-enabled sensors is the input to the LSTM (Long Short-Term Memory) neural networks which are most commonly used to predict traffic conditions shortly [10].

The LSTM-DT based frameworks have been shown to predict urban traffic congestion and vehicle flow patterns more accurately than the conventional approaches like ARIMA and SVR (Support Vector Regression) [11]. Such prediction capabilities are very much beneficial for traffic control that is adaptive and for smart city areas determining the best routing strategies. With the surveys, one of the surveys conducted by Hassija et al. provides a thorough and in-depth description of the technologies, architectures, and challenges that need to be tackled for Digital Twin [12].

They indicate that the three fundamental factors are required for the DT ecosystems to be trustworthy and scalable: interoperability among various systems, secure data exchange, and platform standardization. Moreover, the merging of 5G/6G and edge computing is viewed as the main driver that will enable very low-latency communication in large-scale DT applications, particularly in the case of intelligent transportation systems [13].

Technological advancements that have been made in this area still do not solve all existing problems. The main difficulties that the current Digital Twin (DT) frameworks have to overcome are related to data privacy concerns, real-time synchronization, model validation, and the absence of common standards [14][15]. To solve these challenges interdisciplinary research is needed that mixes IoT systems, cyber-physical modeling, and advanced AI-based analytics. The literature overall suggests that Digital Twin technology can be very useful in the areas of urban traffic management optimization, congestion reduction, and sustainable mobility, which are also the project's objectives.

III. PROPOSED METHODOLOGY

The SUMO traffic simulator will be the primary tool for the development of a virtual urban intersection model consisting of a four-way street with the main objective of giving controlled access to the traffic signals of any direction. The project also intends to build a digital twin framework that integrates real-time simulation through the Traci interface for continuous

supervision and control of the traffic flow and operation.

Another aim of the project is the observation and measurement of the critical traffic parameters, such as the number of vehicles and their average speed, which are necessary for the understanding of traffic behavior and the identification of congestion patterns. Adaptive traffic signal control based on the real-time traffic conditions provided by the system is employed to 'open' signal phases and enhance the productive flow of the traffic. The project will also look into the relationship between traffic congestion and vehicle speed in an attempt to validate the traffic pattern of being urban realistic.

Finally, the project will demonstrate the digital twin technology as an example of the application of intelligent transportation systems and traffic management, optimization, and decision support in smart city environments, thereby illustrating the potential of its real-time applications, etc.

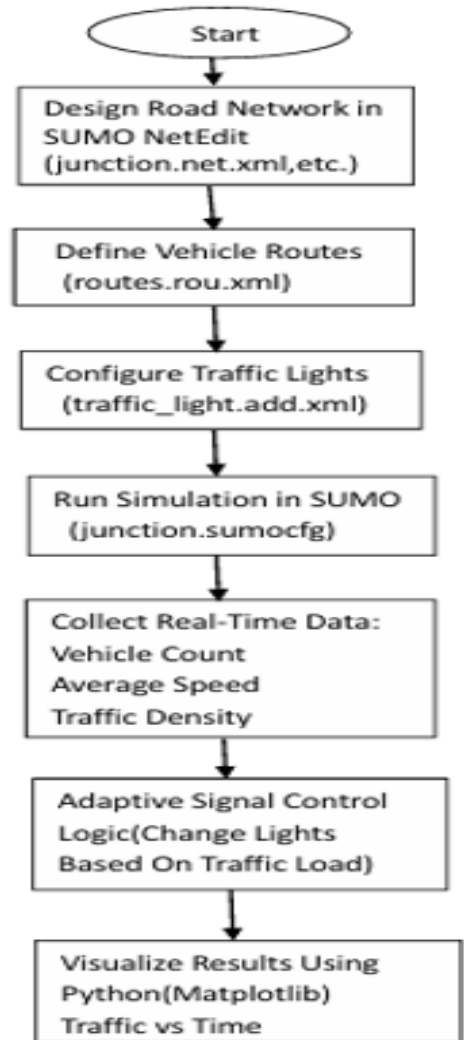


Fig. 1. Working Flow Diagram Of Digital Twin

The authors reveal that the first move toward a digital twin-based traffic management system is the deployment of the SUMO NetEdit tool for the development of the urban

infrastructure for cars. Presently, the intersection's real design is depicted by marking the points, edges, and lanes, which will eventually be written to the network configuration files, like junction.net.xml.

The following step involves the generation of vehicle paths and the management of traffic via the routes.rou.xml file where the traffic coming into and going out of the intersection from all sides will be natural as long as it does not exceed the total traffic on the roads.

After the mapping and the vehicles moving, the light settings come in via the traffic_light.add.xml file which will also specify the different signal phases and timings. The junction.

sumocfg file has combined the road network, vehicle routes, and traffic light settings into a single comprehensive traffic simulation. The TraCI interface collects traffic data in real-time, for instance, the number of vehicles on each street, and keeps on sharing that data during the entire simulation period along with their average speeds and overall traffic density. An adaptive traffic signal control logic that is based on this live data changes the traffic light phases according to the traffic load dynamically. Thus, it contributes to a continuous flow of traffic even during peak congestion.

Real-time traffic data collected is graphically represented through the use of Python libraries like Matplotlib, where charts show the changes in the number of vehicles and their speed over time. The results obtained pointed out that a digital twin has the capability to not only simulate and oversee traffic conditions but also control the traffic flow in real-time.

This research presents a real-time digital twin framework for urban traffic simulation and analysis. The system brings together microscopic traffic simulation, real-time data monitoring, and intelligent control within a single architecture. The system can rapidly respond to the shifts in traffic conditions and alleviate congestion through adaptive signal control by utilizing the real-time vehicle density data.

In addition, the research examines the traffic congestion phenomenon by looking into the correlation between vehicle density and average speed which is very much in line with the real-world urban traffic pattern. The combination of simulation, data analysis, and visualization not only highlights the capability of digital twin technology for smart city traffic management but also for intelligent transportation systems.

The methodology adheres to a well-defined and systematic procedure. The first step involves the SUMO simulator generating a realistic four-way intersection where from all directions vehicles can make a straight, left, or right turn. To mimic the real-world scenario, cars are added according to probability rules, thus making the traffic flow dynamic and non-uniform all through the simulation.

Then, the digital twin is created by interlinking the SUMO simulation with Python via the TraCI interface, which allows for the instantaneous exchange of data between the simulation engine and the control logic. The main traffic parameters

like the number of vehicles on every incoming road and the average vehicle speeds are observed continuously during the simulation steps. The traffic signal control logic is transformed into an adaptable one by applying it according to the levels of congestion detected through the dynamic adjustment of traffic light phases to the quality of the traffic flow. All traffic data is gathered in CSV format and then visualized with Python-based plotting tools for analysis of congestion trends, speed-density relationships, and system performance to be done.

IV. RESULTS AND DISCUSSION

The digital twin-based traffic simulation model that was proposed has been successfully implemented and validated in the context of a four-way urban intersection. By the integration of SUMO and Python via the TraCI interface, the system was able to keep track of various traffic parameters, such as the number of vehicles, traffic density, and their average speed, in real-time. Also, the simulation reports affirmed the method's capability of accurate representation of urban traffic behavior.

Figure 1 illustrates the traffic load over time, which indicates that the total number of vehicles entering the network increased slowly and steadily in the beginning of the simulation period. The increase is due to the probabilistic traffic generation mechanism used for the real demand simulation scenario. After a while, the traffic load could be seen to be gradually stabilizing, indicating that the network had reached the saturation point where the inflow and outflow of vehicles were equal.

This characteristic shows that the intersection and routing logic designed works best and can manage traffic without affecting the system stability. The average speed over time which very clearly illustrated the case of traffic density and vehicle speed being totally opposite to each other. The average speed during low vehicle counts, i.e., the conditions of free-flow traffic, is high. As the number of vehicles increases, congestion is developing, and the average speed drops significantly as a result. This observation is in line with real-world urban traffic patterns where congestion has increased mobility and travel efficiency negatively affected.

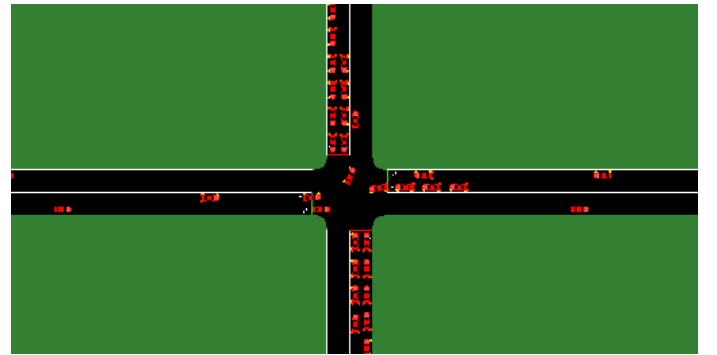


Fig. 2. Simulation Output

according to the predefined routes given in the routes.rou.xml file. Each flow specifies how vehicles

enter the junction, travel, and exit towards their respective destinations. The density and placement of vehicles in all directions confirm the system's successful implementation of bidirectional traffic movement, allowing the vehicles to pass from all sides of the intersection and not just a single lane or route.

The center of the intersection, managed by a traffic light logic (J1), alternates between green and red phases in a dynamic manner, as dictated by the Python-based digital twin controller (digital twin.py). This means the system can mimic real-time signal adjustments corresponding to the number of vehicles getting closer to the junction.

The SIMO interface has at the bottom the message "Simulation started with time: 0.00", which refers to the initialization of the digital twin simulation. The vehicle counter located at the right bottom (205) indicates the total number of vehicles that are presently active in the network and therefore, it is an indicator of real-time traffic flow intensity.

The graph depicted above shows the time-changing nature of the traffic load through the digital twin traffic simulation which was done with the aid of SUMO and Python. The bottom of the graph illustrates the time taken by the simulation in seconds, whereas the top of the graph shows how many vehicles were active in the road network at every second. The line drawn on the graph corresponds to the traffic network where the overall number of vehicles is incremented constantly as the vehicles are coming onto the simulation from several routes that have been defined in the routes.rou.xml file.

To start with, the vehicle count soars at an incredible rate which points to the fact that there is a great inflow rate as the simulation gets going and the traffic is getting at the intersection. Eventually, the slope of the curve gets lesser, which means that the system has attained a steady-state condition at which the number of cars coming in and going out is the same. This basal situation indicates that the digital twin and the traffic signal control have actually been working together to reduce the traffic density equally well in all directions.

The upper section of the diagram exhibits the two primary performance graphs resulting from urban traffic simulation with a digital twin facilitated by SUMO and Python. The uppermost graph represents Traffic Load Over Time, which reflects the total number of vehicles actively present in the network at different times during the simulation, whereas the lower graph shows Average Speed Over Time, which is the overall average speed of all vehicles in the whole network.

The graph went on to show a significant increase in the traffic load during the first 80 seconds, where vehicles from all four directions were constantly flowing into the network. This surge was accompanied by the primary intersection becoming congested due to the overflow of cars. Around the 100s mark, the graph started to show a plateau indicating that the inflow and outflow of vehicles were nearly balanced.

This circumstance of inflow and outflow implies that both

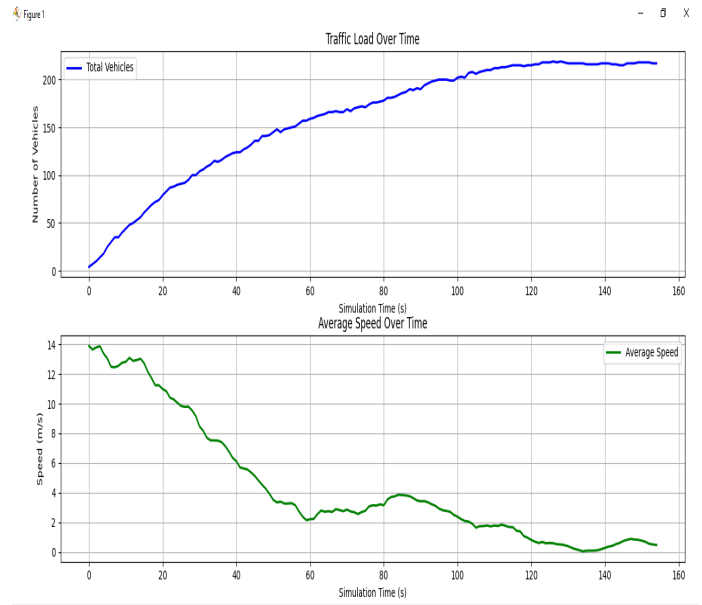


Fig. 3. Traffic Load and Average Speed Analysis

the traffic signal plan and the design of the flows are really effective in maintaining a certain level of traffic density at the intersection.

In the second graph, the pattern of average speed is directly opposite to that of traffic load. At the beginning, when the network is not heavily loaded, cars are able to run at about 13-14 m/s, which is still considered to be smooth traffic flow on the road. However, as the volume of vehicles climbs, the average speed declines and eventually hits its lowest value (almost 0-2 m/s) during the interval of 100-160s when the traffic congestion is the highest. This is exactly the situation of the urban traffic flows where a great number of vehicles results in longer waiting and less movement.

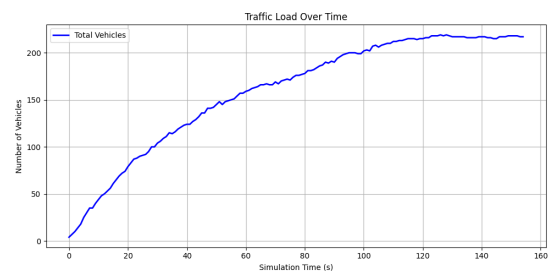


Fig. 4. Traffic Load Analysis

To put it differently, the graph illustrates the urban traffic flow hard to visualize but in retreat and advance terms where congestion is first created and then the control measures make it disappear. Further still, the graph is a confirmation of the model's ability to simulate different traffic and control scenarios. Consequently, it is not only through a representation of the current situation but also as an evaluation tool for the performance of the traffic management system in keeping

a controlled and efficient traffic flow at the intersection in question.

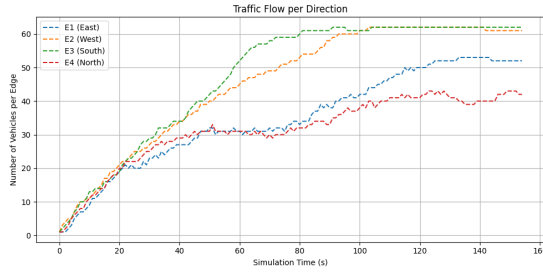


Fig. 5. Traffic Flow per Direction

The diagram illustrates the variations in traffic volume of all four directions throughout the whole simulation. The lines represent the count of vehicles detected on each edge (road) over a period. At the beginning, all the directions experienced an upward trend in vehicle numbers gradually, which indicates normal inflow of traffic in the region. During the simulation, the South (E3) and the West (E2) directions presented more vehicles leading to those roads being either having heavier traffic or waiting longer at the signal. In contrast, the East (E1) and North (E4) roads showed less congestion which meant that the flow was smoother or the signal delays were shorter. The analysis presented here accurately distinguishes the direction with the highest load or congestion and hence is a significant parameter in the design of adaptive traffic light control in digital twin systems. By setting the signal timing based on such data, the overall traffic efficiency and vehicle throughput are going to be improved. In conclusion, the results show that the digit.

V. CONCLUSION

This research work has reached the major milestones that were set up for the creation of a digital twin-based urban traffic simulation system. SUMO was used to create a four-way intersection, and the vehicles were allowed to travel in all directions in accordance with the traffic light signaling. Python combined with SUMO using the TraCI interface allowed the generation of a live digital twin that would monitor the traffic situations such as the number of vehicles and their average speed all the time.

The controlling mechanism for adaptive traffic signals that was put in place responded perfectly to the traffic situation every moment, and hence, the management of traffic flow under different degrees of congestion was improved. The traffic data collected and then visualized made it quite easy to understand that vehicle density and average speed were negatively correlated, thus confirming the simulation environment's realism. All these findings give evidence that the digital twin in question is a mirror of the urban traffic dynamics in the real world.

All in all, the project is a demonstration of how digital twin technology can be applied to intelligent transportation systems. The combining of real-time monitoring, adaptive control, and

data-driven analysis makes it possible for the proposed system to be an urban traffic management solution that is both scalable and efficient. The project outcomes are a clear indication of the digital twins' support of predictive control, optimization, and informed decision-making in the traffic infrastructure of future smart cities.

The project achieved a big success in terms of the development and use of a city traffic management system that was based on Digital Twin and relied on the SUMO simulator, Python, and TraCI integration for the implementation. A very detailed four-way intersection was constructed where vehicles could be allowed to move in all directions while the traffic signal control was actuated dynamically.

The digital twin was able to track the real-time traffic parameters of vehicle density and average speed constantly and change the signal phases accordingly to reduce congestion. The plots indicated a strong inverse relationship between traffic density and vehicle speed, which was an embodiment of the real-world traffic behavior in the simulation.

The proposed system delivered all the objectives stated and proved the role of digital twins in the success of intelligent transportation systems by enhancing real-time monitoring, adaptive control, and data-driven analysis. It is possible that the technological advancement of digital twin has opened up the big potential road towards applied traffic optimization through the means of decision support and smart city mobility planning.

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